

Mathematical Induction (Summation) - Development Questions

- 1 Prove using Mathematical Induction that for all $n \geq 1$, 3

$$1 + 4 + 7 + \cdots + (3n - 2) = \frac{n(3n - 1)}{2}$$

- 2 Prove using Mathematical Induction that for all $n \geq 1$, 3

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2}{4}(n + 1)^2$$

- 3 Verify that for all $n \geq 1$, the sum of the squares of the first $2n$ positive integers are given by the formula: 3

$$1^2 + 2^2 + 3^2 + \cdots + (2n)^2 = \frac{n(2n + 1)(4n + 1)}{3}$$

- 4 Using Mathematical induction, prove that for all positive integers n , 3

$$2 + 5 + 8 + \cdots + (3n - 1) = \frac{n}{2}(3n + 1)$$

- 5 Prove the statement below is true for integers $n \geq 1$. 3

$$1 + 5 + 9 + \cdots + (4n - 3) = 2n^2 - n$$

- 6 Prove the statement below is true for all positive integers. 3

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \cdots + n \times n! = (n + 1)! - 1$$

- 7 Prove the statement below is true for all positive integers. 3

$$1 + x + x^2 + \cdots + x^n = \frac{x^{n+1} - 1}{x - 1}$$

- 8** Prove the statement below is true for all positive integers using mathematical induction. **3**

$$1 + (1 \times 2) + (1 \times 2 \times 3) + \cdots + (1 \times 2 \times 3 \times \cdots \times n) = \frac{1}{2}(n+1)! - 1$$

- 9** Prove the statement below is true for all integers $n \geq 1$ using mathematical induction. **3**

$$1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^{n-1}}$$

- 10** Prove the statement below is true for all integers $n \geq 1$ using mathematical induction. **3**

$$n^3 + (n-1)^3 + (n-2)^3 + \cdots + 1^3 = \frac{n(n+1)}{2} \times \frac{(2n+1)}{3}$$

- 11** Prove the statement below is true for integers $n \geq 1$. **3**

$$(1 \times 3 \times 5) + (3 \times 5 \times 7) + \cdots + (2n-1)(2n+1)(2n+3) = \frac{n(4n^3 + 10n^2 + 8n + 3)}{3}$$

- 12** Prove the statement below is true for integers $n \geq 1$. **3**

$$(1 \times 3) + (2 \times 5) + (3 \times 7) + \cdots + n \times (2n+1) = \frac{n(n+1)(2n+1)}{3}$$

- 13** Prove the statement below is true for all integers $n \geq 1$ using mathematical induction. **3**

$$(1 \times 2 \times 3) + (2 \times 3 \times 4) + \cdots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

- 14** Prove the statement below is true for all integers $n \geq 1$ using mathematical induction. **3**

$$(1 \times 3) + (2 \times 4) + (3 \times 5) + \cdots + n(n + 2) = \frac{n(n + 1)(n + 2)}{3}$$

- 15** Prove the statement below is true for integers $n \geq 1$. **3**

$$1 + 2 + 2^2 + 2^3 + \cdots + 2^n = 2^{n+1} - 1$$

- 16** Prove the statement below is true for all positive integers n . **3**

$$(2 \times 1) + (3 \times 2) + \cdots (n + 1) \times 2^{n-1} = n2^n$$

- 17** Refer to parts (a) and (b) below:

- (a) Show that $\frac{1 - \cos 2x}{2 \sin x} = \sin x$ **1**

- (b) Hence, given n is a positive integer, prove the following statement via Mathematical Induction **3**

$$\sin x + \sin 3x + \cdots + \sin(2n - 1)x = \frac{1 - \cos 2nx}{2 \sin x}$$

- 18** Refer to parts (a) and (b) below:

- (a) Show that $\sin 2nx = \sin[(2n + 1)x] \cos x - \cos[(2n + 1)x] \sin x$ **1**

- (b) Hence, given n is a positive integer, prove the following statement via Mathematical Induction **3**

$$\cos x + \cos 3x + \cdots + \cos(2n - 1)x = \frac{\sin 2nx}{2 \sin x}$$

19 Refer to parts (a) and (b) below:

(a) Show that for $k > 0$, $\frac{1}{(k+1)^2} - \frac{1}{k} + \frac{1}{k+1} < 0$ **1**

(b) Use Mathematical Induction to prove that for all integers $n \geq 2$ **3**

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

20 Prove the statement below is true for integers $n \geq 1$. **3**

$$1 + 2\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right)^2 + \cdots + n\left(\frac{1}{2}\right)^{n-1} = 4 - \frac{n+2}{2^{n-1}}$$